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# Effective Diffusivity and Evaporative Cooling in Convective Drying of Food Material

Chandan Kumar[1], Graeme J. Millar[1], M. A. Karim[1,]

[1]Science and Engineering Faculty, Queensland University of Technology, QLD, Australia

Corresponding author: ,,;. E-mail: azharul.karim@qut.edu.au



## Abstract

This article presents mathematical models to simulate coupled heat and mass transfer during convective drying of food materials using three different effective diffusivities: shrinkage dependent, temperature dependent and average of those two. Engineering simulation software COMSOL Multiphysics was utilized to simulate the model in 2D and 3D. The simulation results were compared with experimental data. It is found that the temperature dependent effective diffusivity model predicts the moisture content more accurately at the initial stage of the drying, whereas, the shrinkage dependent effective diffusivity model is better for the final stage of the drying. The model with shrinkage dependent effective diffusivity shows evaporative cooling phenomena at the initial stage of drying. This phenomenon was investigated and explained. Three dimensional temperature and moisture profiles show that even when the surface is dry, inside of the sample may still contain large amount of moisture. Therefore, drying process should be carefully dealt with otherwise microbial spoilage may start from the centre of the 'dried' food. A parametric investigation has been conducted after the validation of the model.

**KEYWORDS:** Mathematical modelling, evaporative cooling, food drying, effective diffusivity, experimental investigation

## INTRODUCTION

Food drying is a process which involves removing moisture in order to preserve fruits by preventing microbial spoilage. It also reduces packaging and transport cost by reducing weight and volume. Compared to other food preservation methods, dried food has the advantage that it can be stored at ambient conditions. However, not only drying is a very energy intensive process which accounts for up to 15% of all industrial energy usage but also the quality of food may degrade during the drying process [1-3]. The objective of food drying is not only to remove moisture by supplying heat energy but also to produce quality food[4]. To reduce this energy consumption and improve product quality, physical understanding of the drying process is essential. Mathematical models have been proved useful to understand the physical mechanism, to optimize energy efficiency and to improve product quality[5]. Mathematical models could be either empirical or fundamental models. Empirical expressions are common and relatively easy to use[2]. Many empirical models for drying have been developed and applied for different products, for instance, banana[6], apple[7], rice[8], carrot[9], cocoa[10] etc. Erbay et al.[11] reveiewd empirical models for drying, and found that the best fitted model is different for different product. However, these empirical models are only applicable in the range used to collect the experimental parameters[12]. In addition, they typically are not able to describe the physics of drying. In contrast to empirical relationships, fundamental models can capture the physics during drying satisfactorily[13-15]. Fundamental



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mathematical modelling is applicable for a wide range of applications and optimization scenarios[12].

Several fundamental mathematical models have been developed for food drying. For example, Barati .et al[16] developed a food drying model wherein they considered the material properties to be constant. However, in reality during the drying process physical properties such as diffusion coefficients and dimensional changes occur as the extent of drying progresses[17]. Consequently, if these latter issues are not considered, the model predictions may be erroneous in terms of estimating temperature and moisture content[18]. In particular, the diffusion coefficient potentially has a significant effect upon drying kinetics.

The effective diffusivity calculation is crucial for drying models because it is the main parameter that controls the process with higher diffusion coefficient implying increased drying rate. The diffusion coefficient changes during drying due to the effects of sample temperature and moisture content[19]. Alternatively, some authors considered effective diffusivity as a function of shrinkage or moisture content[20], whereas, others postulated it as temperature dependent[21]. In the case of a temperature dependent effective diffusivity value, the diffusivity increases as drying progresses. On the other hand, effective diffusivity decreases with time in the case of shrinkage or moisture dependency. This latter behaviour is ascribed to the diffusion rate decreasing as moisture gradient drops. However, Baini et al.[22] mentioned that shrinkage also tends to reduce the path length for diffusion which results in increased diffusivity. Consequently, there are two opposite



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effects of shrinkage on effective diffusivity which theoretically may cancel each other. Silva et al.[23] analysed the effect of considering constant and variable effective diffusivities in banana drying. They found that the variable effective diffusivity (moisture dependent) is more accurate than the constant effective diffusivity in predicting drying curve. Some authors[20] considered effective diffusivity as a function of moisture content while others[24] considered it as a function of temperature. However, there is limited study comparing the influence of temperature dependent and moisture dependent effective diffusivity. Recently, Silva et al.[25] considered effective diffusivity as a function of both temperature and moisture together (i.e. D=f(T, M)), not temperature or moisture dependent diffusivities separately. Therefore, it was not possible to compare the impact of considering temperature and moisture dependent effective diffusivities. Moreover, they did not report the impact of variable diffusivities on material temperature. The comparison of drying kinetics for both temperature and moisture dependent effective diffusivities can play a vital role in choosing the correct effective diffusivity for modelling purposes. Even though there are several modelling studies of food drying, there are limited studies which compare the impacts of temperature dependent and moisture dependent effective diffusivities.



Understanding exact temperature and moisture distribution in food samples is important in food drying. Joardder et al.[26] showed that the temperature distribution plays a critical role in determining the quality of dried food. Similarly, moisture distribution plays a critical role in food safety and quality. Vadivambal and Jayas[27] showed that despite the fact that the average moisture content was lower than what was considered a safe value,

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spoilage started from the higher moisture content area. Therefore, it is crucial to know the moisture distribution in the sample. Unfortunately, it is difficult to measure temperature and moisture distribution inside the sample experimentally which means that appropriate modelling approaches are required to determine the moisture distribution. Mujumdar et al.[28] argued that technical innovation can be intensified by mathematical modelling which can provide better understanding of the drying process. Karim and Hawlader[20] developed a mathematical model to determine temperature and moisture changes with time but it did not provide temperature and moisture distribution within the sample. The moisture distribution is a key parameter for evaporation because evaporation depends on surface moisture content.



Evaporation plays an important role during drying in terms of heat and mass transfer, with higher evaporation resulting in enhanced drying rates. During the initial stage of drying, the surface is almost saturated, which induces both higher evaporation and moisture removal rates. Due to this higher evaporation rate, the temperature drops at this stage for a short period of time[29, 30]. Recently Golestani et al.[31] also observed reduced temperature in the initial drying phase and they attributed this phenomenon to high enthalpy of water evaporation. The temperature evolution depends upon the heat flux. During drying, two reverse heat fluxes take place; one is inward ‘convection’ heat flux and another is outward ‘evaporative’ heat flux. Again, there are limited studies which have investigated the temperature variation during the initial stage of convection drying based on heat flux.

In this context, the aims of this paper are threefold. Firstly, to develop three drying models based on three effective diffusivities; namely moisture dependent, temperature dependent, and average effective diffusivities. Secondly, to investigate the evaporative cooling phenomena in terms of heat flux. Finally, to conduct a parametric study with validated models.



## MODEL DEVELOPMENT

The model developed in this research considered cylindrical geometry of the food product as shown in Figure 1.

### Governing Equations

Mass transfer equation:

$$\frac{\partial c}{\partial t}+\nabla.\left(-D_{\mathrm{eff}}\nabla c\right)+u.\nabla c=R \tag{1}$$

Where $c$ is moisture concentration, $t$ is time, $D_{\mathrm{eff}}$ is the effective diffusivity, $R$ is the production or consumption of moisture and $u$ is domain velocity.

Heat transfer equation:

$$\rho c_{\mathrm{p}}\frac{\partial T}{\partial t}+\rho c_{\mathrm{p}}u.\nabla T=\nabla.(k\nabla T)+Q_{\mathrm{e}} \tag{2}$$

Where $T$ is temperature at time $t$, $\rho$ is density, $C_{\mathrm{p}}$ is specific heat of material, $k$ is thermal conductivity and $Q_{\mathrm{e}}$ is internal heat source or sink. The heat source term is zero for convection drying but when electromagnetic heating such as microwave is involved then it should be added to the heat transfer equation.

**Initial And Boundary Conditions**

*Initial Condition:*

Initial moisture content, $M_0 = 4\,kg/kg$ dry basis,

Initial temperature $T_0 = 38^0 C$

*Boundary Conditions:*

*Heat transfer boundary conditions:* Both convection and evaporation were considered at the open boundaries. Thus the heat transfer boundary condition was defined by Eq. (3).

$$\boldsymbol{n}.(k\nabla T) = h_T (T_{\mathrm{air}} - T) - h_{\mathrm{m}}\rho (M - M_{\mathrm{e}}) h_{\mathrm{fg}} \tag{3}$$

Where $h_{\mathrm{T}}$ is heat transfer coefficient *(W/m²/K)* , $h_{\mathrm{m}}$ is mass transfer coefficient *(m/s)*, $T_{\mathrm{air}}$ is drying air temperature *($^0C$)*, $M_{\mathrm{e}}$ is equilibrium moisture content *(kg/kg)* dry basis and $h_{\mathrm{fg}}$ is latent heat of evaporation *(J/kg)*.

At symmetry and other boundaries:

$$\boldsymbol{n}.(k\nabla T) = 0 \tag{4}$$

*Mass transfer boundary conditions:*

At open boundaries:

$$\boldsymbol{n}.(D\nabla c) = h_{\mathrm{m}} (c_{\mathrm{b}} - c) \tag{5}$$

Where $c_{\mathrm{b}}$ is bulk moisture concentration.

At symmetry and other boundaries:

$$\boldsymbol{n}.\left(D\nabla c\right)=0 \tag{6}$$

***Variable Thermo Physical Properties***

In food processing, thermo physical properties play an important role in heat and mass transfer simulation[32]. In this simulation, specific heat and thermal conductivity were considered as function of moisture content *($M_w$)* by the following equations[33].

Specific heat, $C_p = 0.811{M_w}^2 - 24.75M_w + 1742$ *(7)*

Thermal conductivity, $K = 0.006M_w + 0.120$ *(8)*

***Effective Diffusivity Calculation***

In this study, three simulations were performed with three different effective diffusivities. The effective diffusivity formulations are discussed below:

*Moisture or shrinkage dependent effective diffusivity:*

Karim & Hawlader[20] presented effective diffusion coefficient as a function of moisture content for product undergoing shrinkage during drying. In this study, the following equation was used to incorporate the shrinkage dependent diffusivity.

$$\frac{D_{ref}}{D_{eff}} = \left(\frac{b_0}{b}\right)^2 \tag{9}$$

Where $D_{ref}$ is reference effective diffusivity which is constant and calculated by slope method from experimental value, $b_0$ and $b$ are the half thickness of the material at time *0* and $t$ respectively.

Thickness ratio obtained by the following equation[34]

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$$b = b_0 \left[ \frac{\rho_w + M_w \rho_s}{\rho_w + M_0 \rho_s} \right] \qquad (10)$$

Where $\rho_w$ is density of water, $\rho_s$ is density of solid.

*Temperature dependent effective diffusivity:*

Temperature dependent diffusivity was obtained from Arrhenius type relationship to the temperature with the following equation[18, 35].

$$D_{eff} = D_0 e^{-\frac{E_a}{R_g T}} \qquad (11)$$

Where $E_a$ is activation energy *(kJ/mol)* , $R_g$ is the universal gas constant *(kJ/mol/K)* and $D_0$ is integration constant *($m^2/s$).*

*Average effective diffusivity:*

The third model considered the average effective diffusivity, $D_{eff\_avg}$ which is average of the temperature and moisture dependent effective diffusivities.

$$D_{eff\_avg} = \frac{\left( D_0 e^{-\frac{E_a}{R_g T}} + D_{ref} \left( \frac{b}{b_0} \right)^2 \right)}{2} \qquad (12)$$

***Heat And Mass Transfer Coefficient Calculation:***

The heat and mass transfer coefficients are calculated from well established co-relations of Nussel and Sherwood number for laminar and turbulent flow over flat plates as shown in Eqs. 13-16. These relationships have been used in drying by many other reserachers[20, 31, 36, 37] and hence justify the use of these relationships.

The average heat transfer coefficient was calculated from Nusselt number (Nu) by using Eqs. (13) and (14) for laminar and turbulent flow, respectively[38].

$$\mathrm{Nu} = \frac{h_{\mathrm{T}} L}{k} = 0.664\,\mathrm{Re}^{0.5}\,\mathrm{Pr}^{0.33} \qquad (13)$$

$$\mathrm{Nu} = \frac{h_{\mathrm{T}} L}{k} = 0.0296\,\mathrm{Re}^{0.5}\,\mathrm{Pr}^{0.33} \qquad (14)$$

Where L is characteristics lengths, Re is Reynolds number and Pr is Prandtl number.

As Fourier's law and Fick's law were similar in mathematical form the analogy was used to find the mass transfer coefficient. Nusselt number and Prandtl number was replaced by Sherwood number (Sh) and Schmidt number (Sc) respectively as in the following relationships:

$$\mathrm{Sh} = \frac{h_{\mathrm{m}} L}{D} = 0.332\,\mathrm{Re}^{0.5}\,Sc^{0.33} \qquad (15)$$

$$\mathrm{Sh} = \frac{h_{\mathrm{m}} L}{D} = 0.0296\,\mathrm{Re}^{0.8}\,Sc^{0.33} \qquad (16)$$

The value of Re, Sc and Pr was calculated by Eqs. (17), (18) and (19), respectively.

$$\mathrm{Re} = \frac{\rho_{\mathrm{a}} v L}{\mu_{\mathrm{a}}} \qquad (17)$$

$$\mathrm{Sc} = \frac{\mu_{\mathrm{a}}}{\rho_{\mathrm{a}} D} \qquad (18)$$

$$\mathrm{Pr} = \frac{C_{\mathrm{pa}} \mu_{\mathrm{a}}}{k_{\mathrm{a}}} \qquad (19)$$

Where $\rho_a$ is density of air, $\mu_a$ is dynamic viscosity of air, $v$ is drying air velocity, $k_a$ is thermal conductivity of air, $C_{pa}$ is specific heat of air. The values of these parameters along with units are presented in Table 1.

**Simulation methodology:**

Simulation was performed by using COMSOL Multyphysics, a finite element based engineering simulation software. The software facilitated all steps in the modelling process including defining geometry, meshing, specifying physics, solving and then visualising the results. COMSOL multiphysics can handle the variable properties which were a function of the independent variables. Therefore, this software was very useful in drying simulation where material properties changed with temperature and moisture content. The simulation methodology and implementation strategy followed in this project is shown in Figure 2. Banana was taken as a sample for this study.

**Input Properties**

Physical properties of banana and other input paratemers used in the simulation programme are listed in Table 1.

## DRYING EXPERIMENTS

Drying tests were performed based on the ASABE (ASABE S448.1) standard. The procedures for ASABE standard are as below:

- Tests should be conducted after drying equipment has reached steady-state conditions. Steady state is achieved when the approaching air stream temperature variation about the set point is less than or equal to $1^0C$.
- The sample should be clean and representative in particle size. It shall be free from broken, cracked, weathered, and immature particles and other materials that are not inherently part of the product. The sample should be a fresh one having its natural moisture content.

The particles in the thin layer should be exposed fully to the airstream.

Air velocity approaching the product should be 0.3 *m/s* or more.

Nearly continuous recording of the sample mass loss during drying is required. The corresponding recording of material temperature (surface or internal) is optional but preferred.

The experiment should continue until the moisture ratio, *MR*, equals 0.05. $M_e$ should be determined experimentally or numerically from established equations.

A tunnel type drying chamber was used in this experiment. The dryer is equipped with a heater, a blower fan and two dampers. Two dampers were used to facilitate the air recirculation and fresh air intake. Both closed loop and open loop tests were possible by



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adjusting the dampers. A temperature controller and blower speed controller was used to maintain constant drying air temperature and air velocity.

The weight of the sample was measured using a load cell, which was calibrated using standard weights. It was understood that the air velocity has considerable effect on the load cell reading and different calibration curves were prepared for different flow velocities through the dryer. The load cell was calibrated after installation in the dryer. Air flow rate was calculated by measuring the air velocity at the entrance of the dying section. A calibrated hot wire anemometer measured the air velocity. T-type thermocouple and humidity transmitter was used to measure the temperature and relative humidity. All the sensors were connected to a data logger to store the information.

For experimental investigation, ripe banana *("Musa acuminate")* of nearly same size was used for drying. First, the banana was peeled and then sliced into 4mm thickness and about 36 mm diameters. Initial moisture content was about 4 *kg/kg* dry basis and final moisture content was between 0.22 to 0.25 *kg/kg* dry basis i.e. moisture ratio was 0.055 to 0.062. Then the slices were put in trays composed of plastic net. Plastic net was used to reduce conduction heat transfer since this effect was neglected in the model. The plastic tray was put into the dryer after reaching steady state condition. Each run included approximately 600 *gms* of material. Following each drying test, the sample was heated at $100^{0}C$ for at least 24 *h* to get bone dry mass.

**UNCERTAINTY ANALYSIS**

Uncertainty analysis of the experiments was done according to Moffat[39]. If the result R of an experiment is calculated from a set of independent variables so that, $R = R\left(X_1, X_2, X_3..........., X_{\mathrm{N}}\right)$. Then the overall uncertainty can be calculated using the following expression

$$\delta R = \left\{ \sum_{i=1}^{N} \left( \frac{\partial R}{\partial X_{\mathrm{i}}} .\delta X_{\mathrm{i}} \right)^2 \right\}^{1/2} \tag{20}$$

and the relative uncertainty can be expressed as follows:

$$e = \frac{\delta R}{R} = \left\{ \sum_{i=1}^{N} \left( \frac{1}{R} . \frac{\partial R}{\partial X_{\mathrm{i}}} .\delta X_{\mathrm{i}} \right)^2 \right\}^{1/2} \tag{21}$$

**Uncertainty analysis of temperature:**

The temperature was directly obtained from the calibrated thermocouple and the accuracy was within the ASHRAE recommended range, which is $\pm 0.5^{0}C$. Therefore, the uncertainty of the temperature would be

$$T = T_{\mathrm{measured}} \pm 0.5 \tag{22}$$

**Uncertainty analysis of moisture content:**

The dry basis moisture content, ratio of the weight of moisture, $W_{\mathrm{m}}$ to that of bone dry weight, $W_d$ , of the sample was calculated from the following equation.

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$$M = \frac{W_{\mathrm{m}}}{W_{\mathrm{d}}} = \frac{W - W_{\mathrm{d}}}{W_{\mathrm{d}}} \tag{23}$$

Therefore, $\delta M = \frac{\partial M}{\partial W}.\delta W + \frac{\partial M}{\partial W_{\mathrm{d}}}.\delta W_{\mathrm{d}} = \frac{\delta W}{W_{\mathrm{d}}} - \frac{W.\delta W_{\mathrm{d}}}{W_{\mathrm{d}}{}^{2}}$ and $\frac{\delta M}{M} = \frac{\delta W}{W - W_{\mathrm{d}}} - \frac{W.\delta W_{\mathrm{d}}}{\left(W - W_{\mathrm{d}}\right).W_{\mathrm{d}}}$

Now the relative uncertainty associated with the measurement of moisture content of sample can be expressed:

$$e_{\mathrm{m}} = \left\{ \left( \frac{\delta W}{W - W_{\mathrm{d}}} \right)^{2} + \left( \frac{W.\delta W_{\mathrm{d}}}{\left(W - W_{\mathrm{d}}\right).W_{\mathrm{d}}} \right)^{2} \right\}^{1/2} \tag{24}$$

Now the present work considers following valued of the banana sample to be dried in the drying chamber. $W = 600g$ and $W_{\mathrm{d}} = 120g$ .As these two values are obtained using the same load cell, and as per manufacturer's specification, the percentage error of load cell is $\pm 0.1\%$, therefore, $\delta W = \delta W_{\mathrm{d}} = 0.0001$. Substituting all the valued in Eq. 24, the relative uncertainty for moisture content, $e_{\mathrm{m}}$, is obtained, the value is found to be $\pm 1.06\%$ .

## RESULTS AND DISCUSSION

The validation of the model was done by comparing the moisture and temperature profile obtained from experiment and simulation. Fig. 3 represents the comparison of moisture profile obtained by experiments and models considering three different effective diffusivities. Results show that simulated moisture content with temperature dependent diffusivity closely agreed with the experimental moisture data at the initial stage of the drying process. On the other hand, the shrinkage dependent diffusivity model exhibited a faster drying rate at the initial stage but followed experimental data closely at the final stage of drying. This higher drying rate during the initial stage can be attributed to the higher diffusion coefficient in that stage. Moisture dependent effective diffusivity is higher at the initial stage as can be seen from Eqs. (9) and (10). These two equations show that initially the diffusivity value was greater at higher moisture content and then decreased with moisture content. Golestani et al.[31] also found higher drying rate compared to the experimental results for both models obtained from two effective diffusivities with and without shrinkage. Therefore, more complex and physics based formulation is necessary to calculate effective diffusivity and predict the moisture content more accurately. However, consideration of effective diffusivity as an average of those two effective diffusivities provided a better match with experimental data. Similar result was found by the study of Golestani et al.[31].

The temperature profile of the material is shown in Fig. 4 for drying air temperature $60^0C$ and velocity 0.5 *m/s*. The predicted temperatures agreed reasonably well with the experimental data. However, interestingly for the shrinkage dependent effective diffusivity model there was an drop in temperature at the beginning of the drying process. This was probably due to the evaporative cooling of the product. At the initial stage of drying, the surface of the sample was saturated with moisture and evaporation rate was higher. Thus evaporative heat was taken away from the material resulting in a temperature drop. The increased evaporation (higher drying rate) can also be seen in Fig. 3 for the shrinkage dependent effective diffusivity curve. For better visualization, the temperature profile was plotted for small time steps in Fig. 5, wherein temperature reduction was noted for the first few minutes of drying. A decreasing temperature profile at initial stage of drying was also obtained by Turner and Jolly[30] and Zhang and Mujumdar[29] in microwave convective drying and Golestani et al.[31] in convection drying in their simulation. However, they just reported these results without any interpretation for this event. To investigate this observation further, the inward heat flux, outward heat flux and total heat flux were plotted in a single graph as shown in Fig. 6. The inward heat flux was due to convection (from air to material) and outward heat flux was due to evaporation (from material to air). Fig. 6 showed that for first 15 minutes of drying the total heat flux was negative due to evaporation which caused the temperature drop in the product. This phenomenon is important in food drying where increase of temperature can cause quality degradation. If this mechanism of cooling could be sustained longer, then the quality of dried food may be improved. Sometimes intermittent drying can be executed to get more evaporation when drying resumes after each tempering period.

More experimentation with continuous temperature measurement should be undertaken to further validate this phenomenon.

As outlined above, temperature and moisture distribution in the food at any instance is important because spoilage can start from higher moisture content region. Sometimes the centre may have higher moisture content though the surface is already dried or with equilibrium moisture content. Consequently, investigating the temperature and moisture distribution is critical in the case of food drying. The modelling and simulation study was helpful in this regard, since it was difficult to measure the moisture distribution



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experimentally. Fig. 7 shows three dimensional temperature and moisture distribution after 40 minutes of drying. It is interesting that, although the surface moisture content ultimately became 0.2 *kg/kg* db, the centre contained 0.6 *kg/kg* db moisture (Fig. 7a). Similar moisture profiles were obtained by Perussello et al.[37]. Although, the drying process may appear to be visually complete, spoilage or microbial growth could still initiate from the moist central region. Therefore, the difficulty in removing moisture from the product centre is one major disadvantage of convective drying.

In regard to temperature distribution, Fig. 7b indicated that the temperature gradient was not significant inside the material because the thickness of the material was very small in the simulation.

## PARAMETRIC STUDY

Parametric study was important to examine the effect of various process parameters on drying kinetics. After the validation of the model, a parametric analysis was conducted in COMSOL Multiphysics. Fig. 8 illustrates the effect of drying air temperature on drying curve at constant air velocity of 0.7*m/s*. It was clear from Fig. 8 that the increase in drying air temperature greatly increased the drying rate. For example, it took 500, 300 and 200 minutes to reach a moisture content value of 0.75*kg/kg* dry basis at drying air temperature of 40, 50 and $60^0C$, respectively. However, the elevated drying air temperature can decrease the product quality (*e.g.* nutrients). Therefore, the drying process has to be optimized and product quality should be investigated along with drying kinetics.

Fig. 9 shows the drying curve for different air velocities. It is evident that increasing drying air velocity increased the drying rate but the effect was not as significant as the effect of temperature. This is because, in convective drying, drying is dominated by internal diffusion. As the drying rate is very high at the beginning, no constant drying rate period is evident. The surface becomes dry quickly and the increasing the velocity does not affect the evaporation because there is not sufficient moisture accumulated on the surface. Therefore, the velocity increase has no effect on the drying rate. These finding conforms with the drying rate curves presented by Karim and Hawlader[20] showing that the drying rate is significantly different for temperature difference whereas it is almost same for velocity changes.

## CONCLUSIONS

In this study, three simulation models have been developed based on three different effective diffusivities. The models were validated with experimental results. Variable material properties were considered in the simulation. Temperature dependent effective diffusivity model predicted the initial stage of drying accurately, whereas moisture dependent effective diffusivity simulations predicted the final stage well. The evaporative cooling phenomena which occurred during the initial stage of drying was, for the first time, investigated and explained. This observation may have significant implication in regards to product quality improvement. Further research to verify this latter phenomenon experimentally may lead to better fundamental understanding and ultimately be applied to limit product temperature to ensure higher product quality. Three dimensional temperature and moisture distribution were presented. The three dimensional graphs suggested that although the surface of the product was dry, the centre moisture content was significant. Parametric analysis showed that by increasing drying air temperature, drying rate can be significantly improved. However, drying air velocity (flow rate) has negligible impact on drying rate.



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**Table 1:** Input conditions for modelling studies

| **Properties** | **Value (unit)** | **Reference** |
|---|---|---|
| Density of Banana, $\rho$ | $980\left(\frac{kg}{m^3}\right)$ | [20] |
| Initial Moisture Content (dry basis), | $4\left(\frac{kg}{kg}\right)$ | Measured |
| Latent heat of Evaporation, $h_{fg}$ | $2358600\left(\frac{J}{kg}\right)$ | [40] |
| Thermal Conductivity of air, $k_{air}$ | $0.0287\left(\frac{W}{mK}\right)$ | [40] |
| Density of Water , $\rho_w$ | $994.59\left(\frac{kg}{m^3}\right)$ | [40] |
| Dynamic viscosity of air, $\mu_{air}$ | $1.78x10^{-4}\left(Pa.s\right)$ | [40] |
| Specific heat of air, $C_{pa}$ | $1005.04\left(\frac{J}{kgK}\right)$ | [40] |
| Density of air, $\rho_a$ | $1.073\left(\frac{kg}{m^3}\right)$ | [40] |
| Equilibrium moisture content, $M_e$ | $0.29\left(\frac{kg}{kg}\right)$ | [41] |
| Specific heat of water, $C_{pw}$ | $4184\left(\frac{J}{kg}\right)$ | [40] |
| Diffusion coefficient, $D$ | $2.41x10^{-10}\left(\frac{m^2}{s}\right)$ | [20] |



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**Fig. 1**. (a) Actual geometry of the sample slice and (b) Simplified 2D axisymmetric model domain.

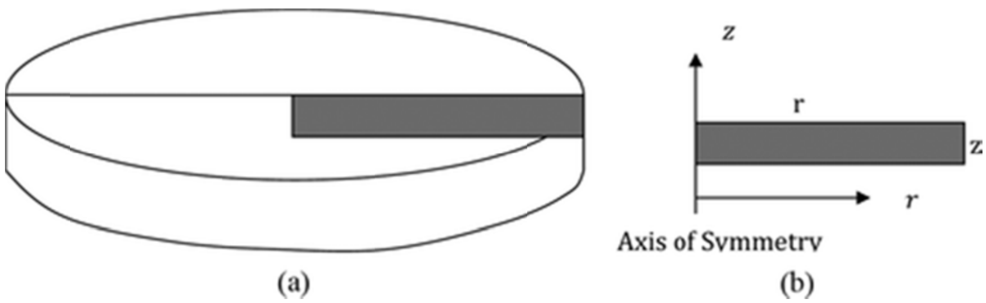




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**Fig. 2.** Simulation strategy in COMSOL multiphysics

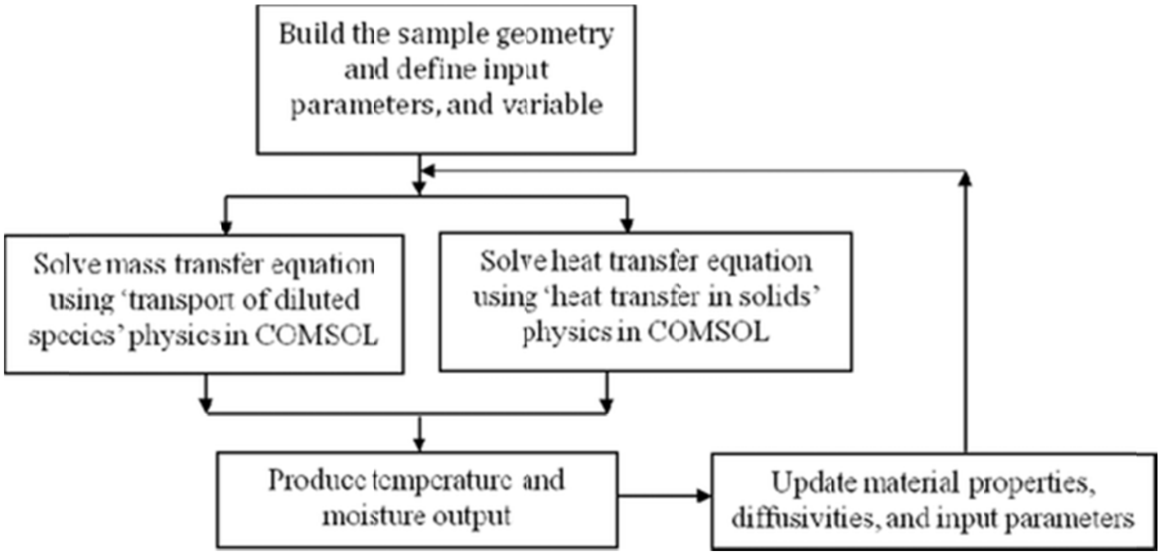

**Fig. 3.** Moisture profile obtained for experimental and simulation with shrinkage and temperature dependent diffusivities ($T$=60$^0$C and $V$=0.7$m/s$).

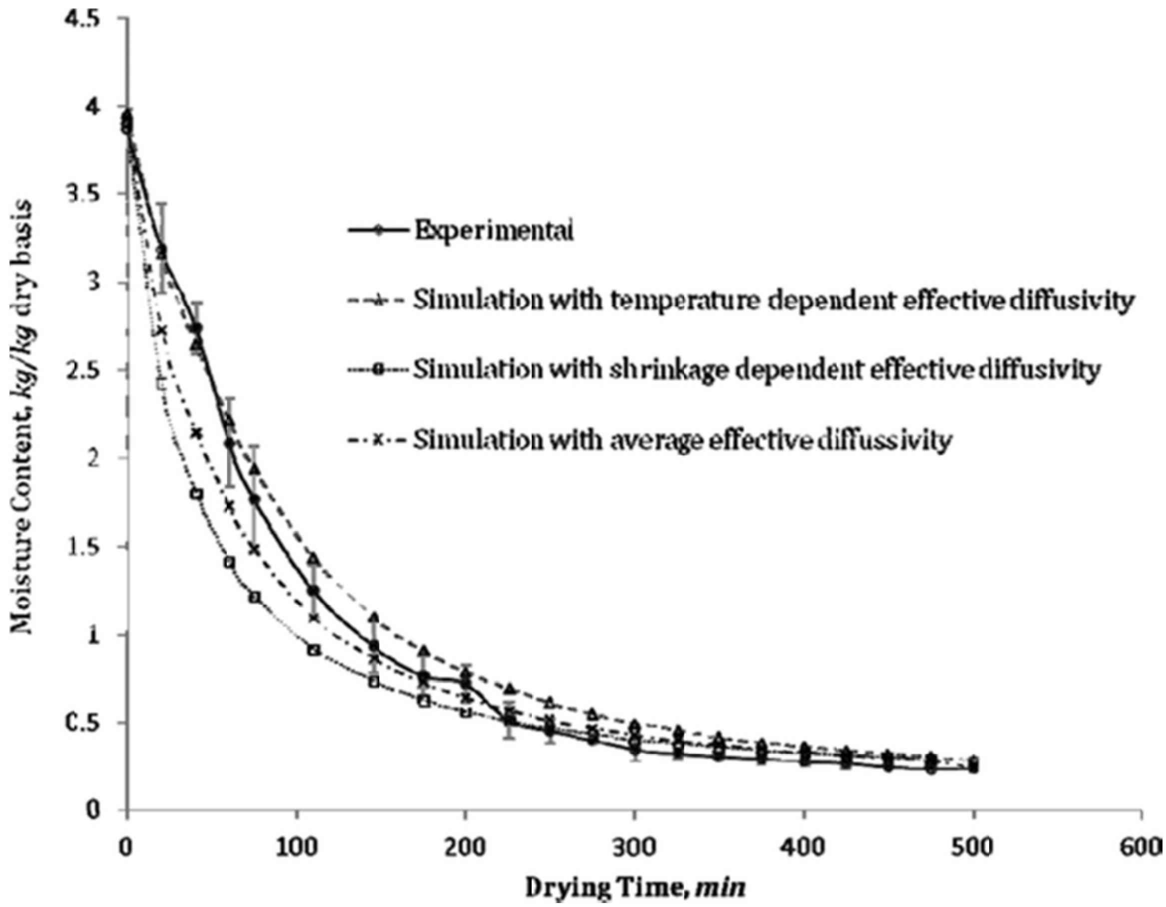

**Fig. 4.** Temperature profile obtained for experimental and simulation with shrinkage and temperature dependent diffusivities (for $T$=60$^0$C and $V$=0.5 *m/s*)

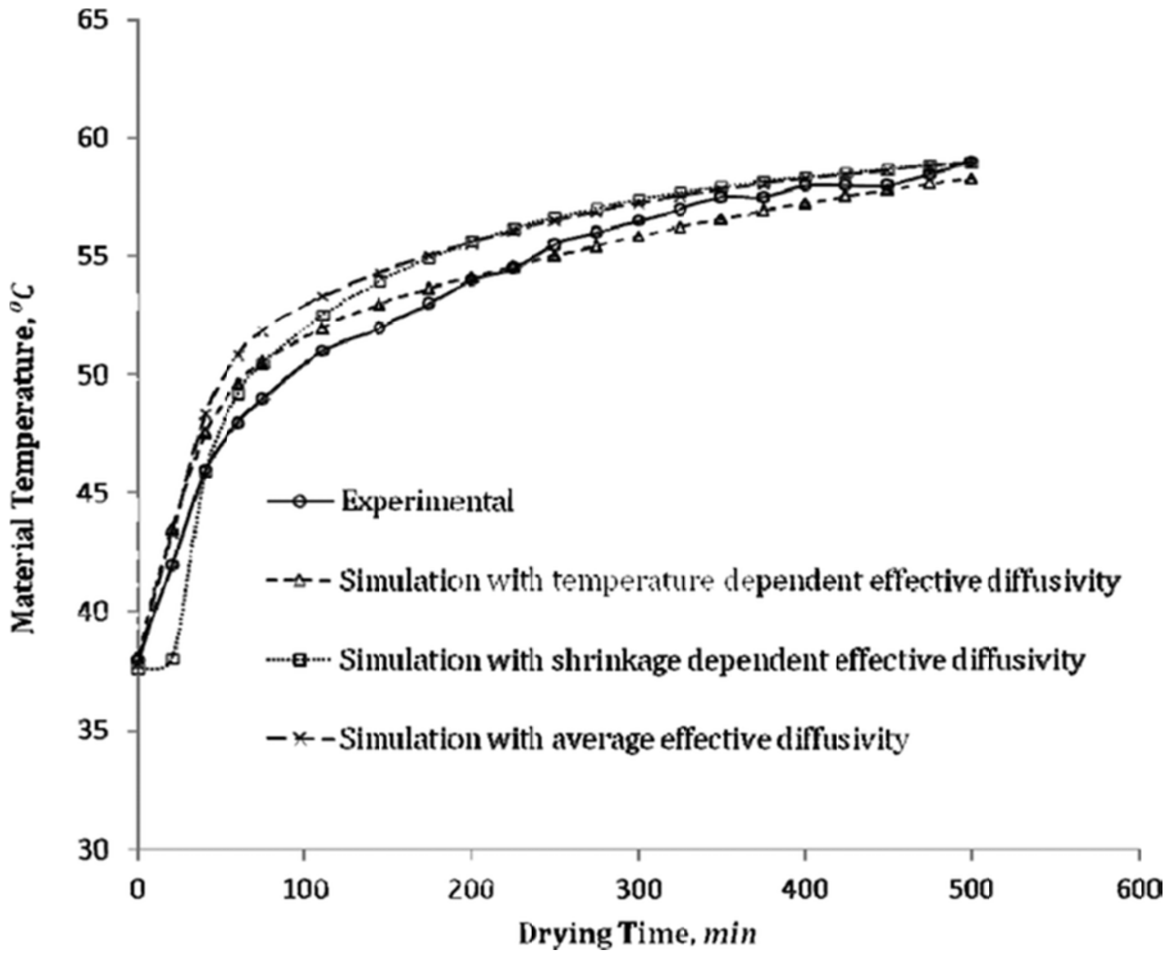

**Fig. 5.** Temperature curve from simulation for shrinkage dependent diffusivity ($T$=60$^0C$ and $V$=0. 5$m/s$)

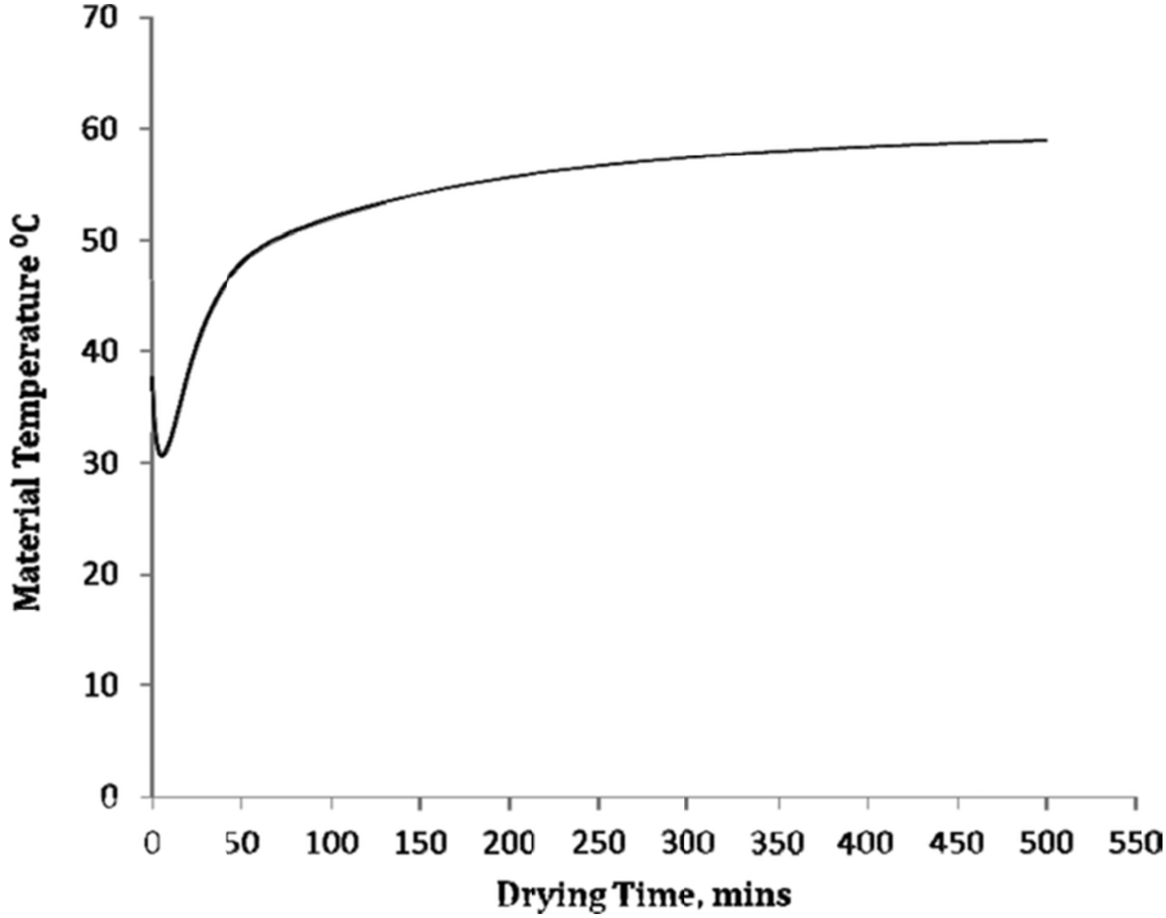




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**Fig. 6.** Evolution of inward (convective), outward (evaporative) and total (convective+evaporative) heat flux

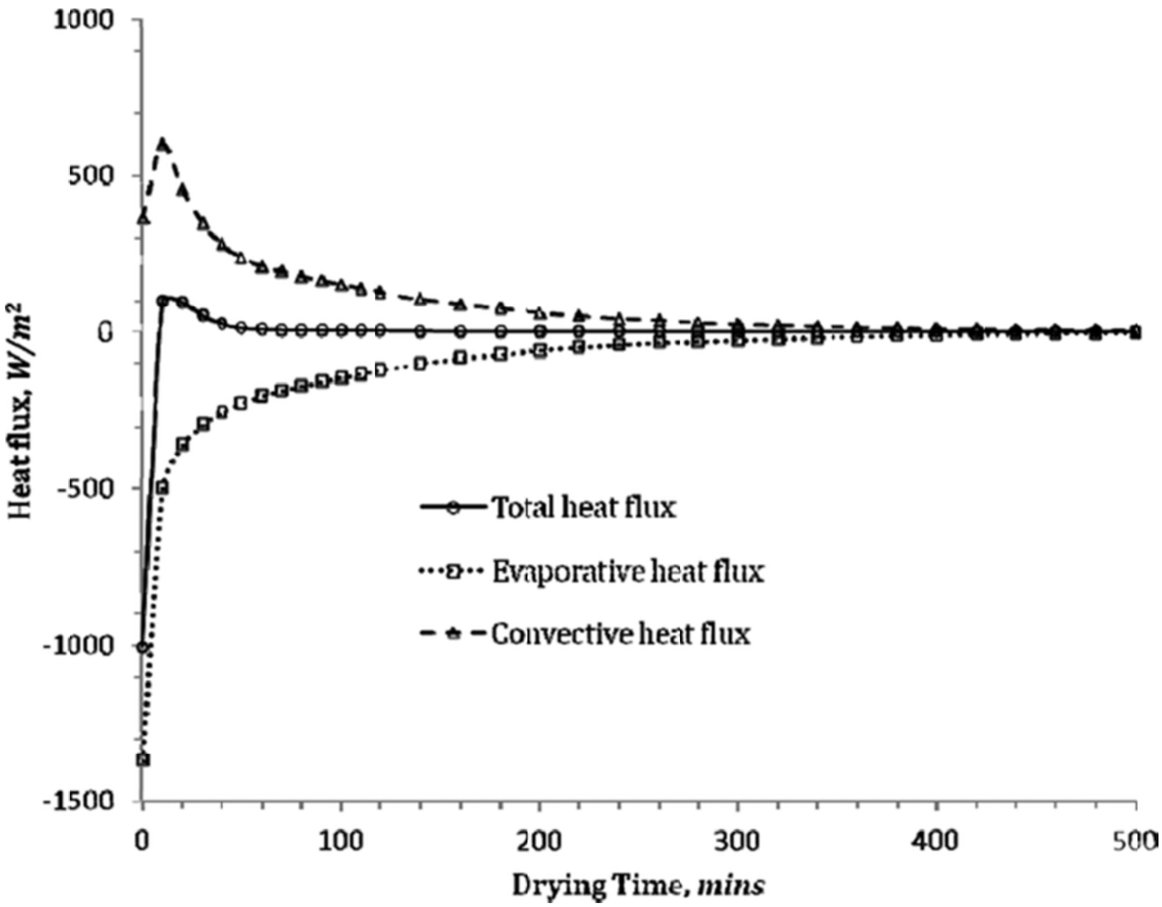

**Fig. 7.** (a) Moisture and (b) temperature distribution in the food after 40 minutes of drying at $T$=60$^0$$C$ and $V$=0.7$m/s$.

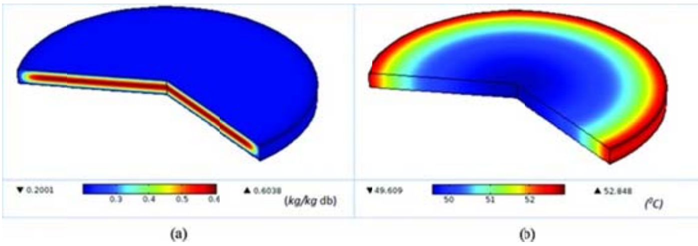




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**Figure 8.** Moisture content for different air temperature for velocity 0.7*m/s*

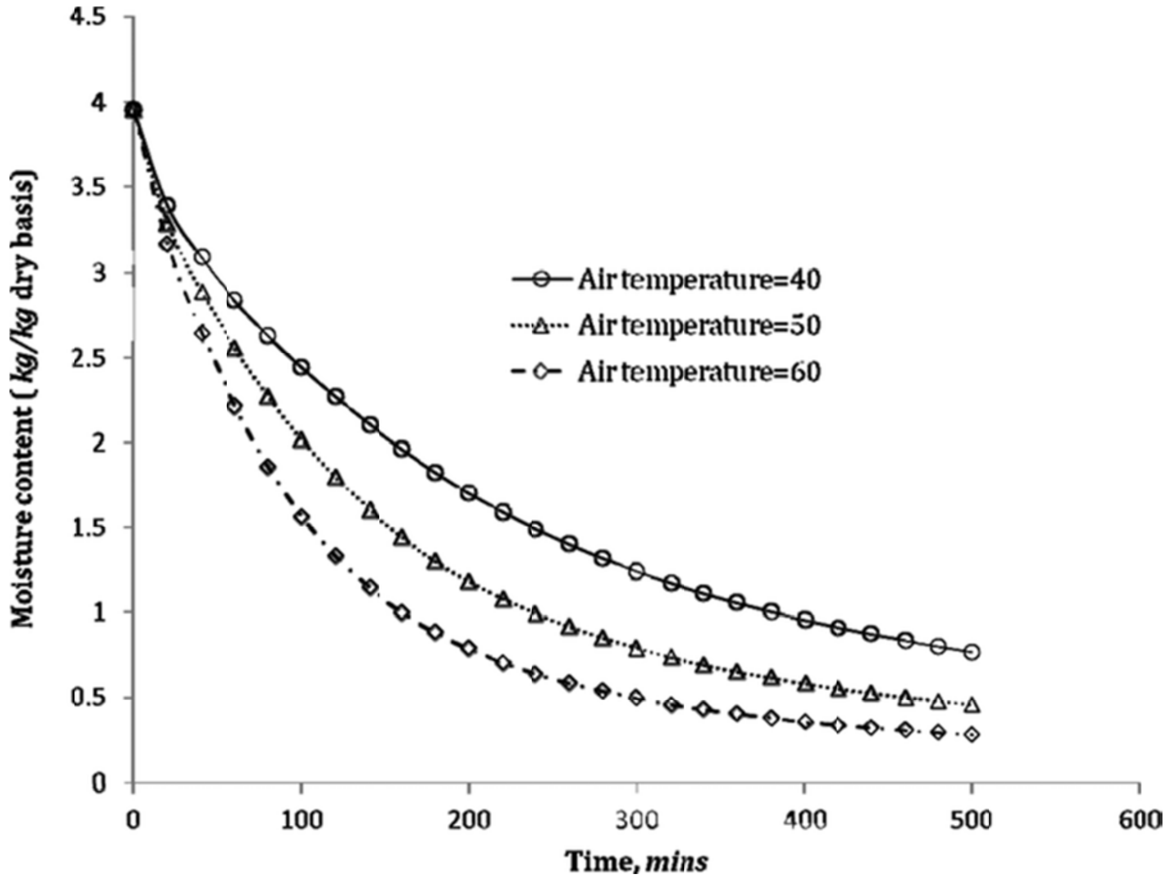

**Figure 9.** Moisture content for different air velocity at $60^{0}C$